LOAD: Load-Aware Dual-Price GNN Routing for Cross-Layer Secure Relay in Multi-Layer NTN

Hyeonsu Lyu, *Member, IEEE*, Hyeonho Noh, *Member, IEEE*,
Hyun Jong Yang, *Senior Member, IEEE*, and Kaushik Chowdhury, *Fellow, IEEE*

***Abstract*—As a key enabler of borderless and ubiquitous connectivity, multi-layer non-terrestrial networks (NTNs) are expected to be a cornerstone of 6G wireless communications. However, the multi-tiered and global-scale nature of these networks amplifies security vulnerabilities due to hidden eavesdroppers whose channels and locations cannot be observed. This paper introduces a joint optimization framework for multi-hop relaying in multi-layer NTNs that maximizes the minimum user throughput while ensuring that the probability of a strictly positive secrecy capacity (SPSC) remains above a prescribed threshold. We derive a tractable closed-form SPSC approximation under cooperative jamming, and an $\mathcal{O}(1)$ globally optimal frequency allocation and transmit–jamming power-splitting strategy for any fixed relay topology, reducing the joint radio-resource-and-routing problem to minimax relay-load routing. We then introduce LOAD, a self-supervised graph neural network (GNN) that predicts prospective relay prices to guide feasible path selection without optimal-route or dual-price labels. LOAD improves mean max-min throughput over baseline routing schemes by up to 2.15 kbps at a 10^{-3} SPSC outage target, tracking max-min secrecy throughput within 3%. We further construct position-data-driven testbeds from satellite, maritime, and cellular deployment data and validate LOAD under geographically constrained relay opportunities and spatially heterogeneous security conditions, demonstrating its practical effectiveness in deployment-informed network settings. The dataset and supplementary material are available at https://github.com/hslyu/sagsin-dataset.**

***Index Terms*—Non-terrestrial network, multi-hop relay, graph neural network, max-min throughput.**

I. Introduction

Non-terrestrial networks extend connectivity to underserved regions by complementing terrestrial infrastructure with low-Earth-orbit (LEO) satellites, high-altitude platforms (HAPs), aerial, and maritime platforms, forming space–air–ground–sea integrated networks (SAGSINs) [1], [2]. Given that the source and the user are often separated by distances far exceeding the range of a single link, end-to-end connectivity across these layers typically relies on multi-hop relaying. However, the broad wireless coverage and predominantly line-of-sight (LoS) propagation characteristic of SAGSIN links substantially increase the risk of eavesdropping [2]–[4]. Moreover, the passive and geographically dispersed nature of Eve'seavesdroppers (Eves) makes their instantaneous channels and locations difficult to acquire in large-scale SAGSINs. Accordingly, reliable multi-hop connectivity in SAGSINs requires the links to maintain a guaranteed level of secrecy, which becomes challenging to sustain when relay resources are shared among multiple competing flows.

This creates a fundamental cross-layer challenge since security requirements introduce resource overhead that propagates through shared relays, thereby affecting the network-wide throughput. Cooperative jamming, for example, improves secrecy against wiretapping but reduces the power available for data transmission [5], [6]. The resulting loss in link efficiency increases the bandwidth required by the flows traversing the corresponding relays, potentially creating network bottlenecks. Secure routing must jointly account for link secrecy feasibility and the network-wide load required to secure relay routes.

Prior work has mainly treated physical-layer secrecy (PLS) and radio resource management (RRM) as separate design problems. PLS studies optimize secrecy performance for predefined link or relay configurations, whereas statistical secure-routing studies select individual source–destination paths under fixed or preassigned RRM [7]–[19]. Conversely, multi-hop RRM studies optimize routing and resource sharing while taking link security feasibility and efficiency as given [20], [21]. However, how statistical eavesdropping risk interacts with network topology and shared relay resources to determine network utility remains unsolved.

This gap motivates the following research question:

How much throughput does a SAGSIN sacrifice to guarantee secrecy against hidden Eves?

We consider passive Eves whose instantaneous channels and locations are unknown but whose spatial distributions can be statistically characterized. We formulate multi-user multi-hop routing as a max-min throughput problem subject to a prescribed strictly positive secure connection (SPSC) probability on each hop. The formulation jointly optimizes relay topology, bandwidth allocation, and transmit–jamming power without instantaneous Eve's channel state information (CSI). Our numerical results further show that max-min throughput closely tracks max-min secrecy throughput under the stringent SPSC targets relevant to secure operation.

Routing determines the required jamming and bandwidth, while the resulting spectral efficiencies and relay loads determine which routes are desirable. To make this coupling tractable, we derive a closed-form SPSC approximation that converts Eve statistics into SPSC-feasible links and jamming

Hyeonsu Lyu and Hyun Jong Yang are with the Institute of New Media and Communications, Seoul National University, Korea (e-mail: {hs.lyu, hjyang}@snu.ac.kr). Hyun Jong Yang is also with the Department of Electrical and Computer Engineering, Seoul National University, Korea. Hyeonho Noh is with Department of Information & Communication Engineering, Hanbat National University, Korea (email: hhnoh@hanbat.ac.kr). Kaushik Chowdhury is a Chandra Family Endowed Distinguished Professor in the Department of Electrical and Computer Engineering, University of Texas at Austin, USA (email: kaushik@utexas.edu).

requirements. For any feasible topology, the optimal bandwidth and transmit–jamming power allocation are obtained in closed form, reducing the original cross-layer problem to minimax normalized relay-load routing. Hidden Eves consequently reduce throughput through two mechanisms: they remove links that cannot satisfy the SPSC target and increase relay loads through the jamming required on the remaining links.

The remaining routing problem is combinatorial even after eliminating the RRM variables, since each user's path selection affects the routing decisions for subsequent users. We therefore propose load-aware dual-price routing (LOAD), which predicts prospective relay prices from the partial topology and the demand of unconnected users. LOAD combines these prices with analytical load increments and fixed-hop shortest-path decoding to construct SPSC-feasible routes. It learns only the future bottleneck information unavailable to the analytical model, while security feasibility, RRM, and path decoding remain model-based. Its self-supervised training requires neither globally optimal routes nor ground-truth dual-price labels.

The contributions of this work are summarized as follows:

- ***Cross-layer secure-relaying formulation***: We formulate multi-user, multi-hop relaying under hidden Eves as a max-min throughput problem with an SPSC target, in which cooperative jamming propagates through shared relays as increased load. This formulation isolates the two mechanisms through which hidden Eves reduce throughput—infeasible-link removal and residual-link jamming overhead—and we further demonstrate that max-min throughput provides a practically accurate surrogate for max-min secrecy throughput in the stringent-security regime.
- ***Statistical-security reduction***: We derive a tractable closed-form SPSC approximation under cooperative jamming, accurate to within 1.6×10^{-3} of Monte Carlo estimates. This enables accurate calculation of the required jamming power for a given SPSC threshold when Eve's CSI is unknown. In turn, the probabilistic SPSC constraint is transformed into deterministic jamming power and routing topology constraints, thereby integrating physical-layer security into the cross-layer resource-allocation and routing framework.
- ***Optimal RRM-assisted problem reduction***: For each feasible topology, we prove that the optimal bandwidth allocation and transmit–jamming power split are obtained in $\mathcal{O}(1)$ time and are globally optimal, which eliminates the continuous RRM variables and reduces the joint problem exactly to minimax relay-load routing—a combinatorial problem over routing topologies alone.
- ***Load-aware dual-price routing***: LOAD exploits the dual structure of the minimax relay-load problem by using a graph neural network (GNN) to learn only the prospective bottleneck price, which is originally unavailable to the primal problem. Its self-supervised training requires neither optimal-route nor dual-price labels, and LOAD achieves a max-min throughput within 3% of the Exhaustive search.

II. Related Works

We survey recent physical-layer security and secure routing approaches in SAGSINs, summarized in Table I.

TABLE I
Summary of the Related Works in Sec. II

Ref.	Archit.	Eve CSI	RA	PC	Jam.	# Hops	LC	Solver
[3]	SAGIN	Known	✗	✗	✗	Two	✗	Analytic
[7]–[9]	SAGIN	Known	✗	O	O	Two	✗	CVX
[4]	SAGIN	Known	✗	O	✗	–	✗	DL
[10]	SAGIN	Known	✗	O	✗	Two	✗	DRL
[22],[23]	SAGIN	Unknown	✗	✗	✗	Multi	✗	Heuristic
[24]	Aerial	Unknown	✗	✗	✗	Multi	✗	Heuristic
[11]	Aerial	Known	✗	✗	O	Multi	✗	Heuristic
[12]	SAGIN	Known	✗	✗	O	Multi	✗	Analytic
[13]	SAGIN	Known	✗	✗	✗	Multi	✗	Analytic
Ours	SAGSIN	Unknown	O	O	O	Multi	O	Analytic+GNN

RA: Frequency resource allocation, PC: Power control, Jam.: Jamming, # Hops: Number of relay hops, LC: Inter-user load-coupled routing

A. Secure Communications in SAGSINs

[3] proposes an aerial bridge scheme using unmanned aerial vehicles (UAVs) to form secure tunnels and reduce the eavesdropping footprint. [7] enhances secrecy rate via a joint UAV deployment and power allocation using successive convex approximation to decouple complex multi-layer constraints, and [8] further enhances security by optimizing UAV trajectory and power with a hybrid free-space optical/radio-frequency (FSO/RF) scheme that mitigates radio-frequency vulnerabilities through optical satellite links. Similarly, [9] combines UAV-assisted non-orthogonal multiple access with cooperative jamming to counter both internal and external eavesdropping threats in SAGINs. [4] proposes a label-free deep learning framework that dynamically selects access strategies to maximize sum secrecy rate. [10] optimizes HAP trajectories and beamforming by employing a generative artificial-intelligence-based DRL framework for proactive adaptation and improved secrecy energy efficiency under dynamic channel changes.

These studies improve secrecy performance through system configuration and PLS optimization, but they consider prescribed connectivity or at most two-hop relaying with known Eve CSI. However, such assumptions do not fit wide-area SAGSINs, which require secure multi-hop routing without instantaneous Eve CSI.

B. Secure Routing in NTNs

[22] presents hierarchical control for secure and quality-of-service-aware routing in SAGINs, and [23] designs an artificial bee colony routing protocol for UAV networks. [24] develops position-aware routing against wormhole and blackhole attacks in airborne mesh networks. These works advance protocol-level path selection without modeling the aggregate resource loads created when multiple user routes share relays.

Physical-layer relay studies consider complementary security mechanisms. [11] jointly designs collaborative beamforming and UAV operation for secure, energy-efficient relaying. [12] analyzes satellite–terrestrial relaying with a friendly jammer, and [13] studies secure relay selection and user scheduling in hybrid satellite–terrestrial networks. These studies analyze secure relay configurations for given links but do not construct a shared multi-user routing tree.

The closest line of research is statistical secrecy-outage-aware routing with jammers [17]–[19]. These studies incorporate statistical eavesdropping risk into individual source–destination path selection under fixed radio resources. Their path evaluation therefore does not capture how resource consumption changes shared-relay loads and the routing opportunities of subsequent users.

The remaining gap is load-coupled multi-user routing under SPSC constraints. Multiple user paths share relay bandwidth and power, so each committed path changes relay loads, may shift the network bottleneck, and affects subsequent routing decisions. SPSC constraints strengthen this coupling by removing insecure links and determining the jamming power and spectral efficiency of the remaining links. The surveyed methods do not jointly optimize such routes with frequency allocation and transmit–jamming power under unknown Eve CSI. This gap motivates the cross-layer formulation and prospective relay-price prediction developed in this paper.

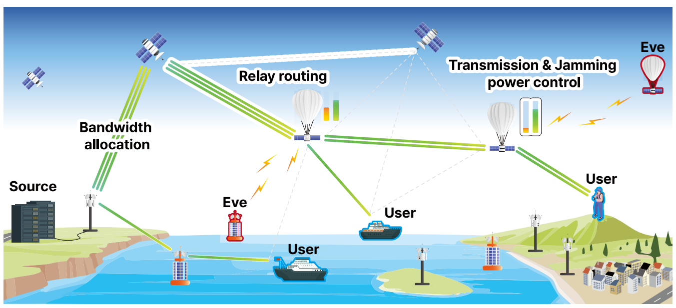


Fig. 1. Target SAGSIN scenario where a source securely relays data to multiple users through intermediate nodes.

III. System Model

A. Scenario Description

This paper aims to construct a secure relaying route in the SAGSIN, with consideration of radio resources, keeping the SPSC probability above a certain threshold in the presence of the hidden Eves. Fig. 1 illustrates the target scenario with three control variables. The SAGSIN deploys decode-and-forward relay where throughput is inversely proportional to the number of relay hops [25]. The nodes transmit predefined jamming signals with the data signal to degrade Eve's wiretap ability by reducing the signal-to-interference-plus-noise ratio (SINR) of Eve's channel [5]. Then, legitimate nodes can recover the original data by decoding the known jamming patterns [26].

The index set of Eves is defined as $\mathcal{M}$. Eves are present at each network layer to wiretap legitimate communications. As the locations of Eves are unknown, their locations are modeled by independent homogeneous Poisson point processes (HPPPs) with densities λ_1, λ_2, λ_3, and λ_4 for the maritime, ground, HAP, and LEO layers, respectively.

We define the index set of nodes as $\mathcal{I} = \{0, 1, ..., I\}$. A total of U users are served through the multi-hop relays, where the user index set is denoted as $\mathcal{U} = \{1, ..., U\}$. The binary variable $x_{(i,j)} \in \{0, 1\}$ indicates whether there exists a hop between the i-th and j-th nodes. The location of the i-th node and Eve e are denoted as $\boldsymbol{p}_i$ and $\boldsymbol{p}_e$, respectively. All notations and variables are summarized in Table III on Appendix A in the supplementary material.

B. Propagation Model and Secrecy Metrics

The instantaneous received signal-to-noise ratio (SNR) of the legitimate hop between node i and j, if any, is given by

$$\mathrm{SNR}^{\mathsf{s}}_{(i,j)} = \frac{\rho_i G_{(i,j)} |h^{\mathsf{s}}_{(i,j)}|^2}{n_0 (d^{\mathsf{s}}_{(i,j)})^{\alpha_i}}, \tag{1}$$

where $G_{(i,j)}$ is the antenna gain, $d^{\mathsf{s}}_{(i,j)} = \|\boldsymbol{p}_i - \boldsymbol{p}_j\|_2$ is a distance between; ρ_i is the power density, $h_{(i,j)}$ is small-scale fading, and n_0 is the noise spectral density. The path loss exponent α_i is determined based on the network layer in which node i is located. Meanwhile, the SNR of wiretap link for node i and Eve k can be defined as

$$\mathrm{SNR}^{\mathsf{e}}_{(i,e)} = \frac{\rho_i G_{(i,e)} |h^{\mathsf{e}}_{(i,e)}|^2 (d^{\mathsf{e}}_{(i,e)})^{-\alpha_i}}{\sigma_i G_{(i,e)} |h^{\mathsf{e}}_{(i,e)}|^2 (d^{\mathsf{e}}_{(i,e)})^{-\alpha_i} + n_0}, \tag{2}$$

where σ_i is the power spectral density of the transmitted jamming signal [27]–[29] and $h^{\mathsf{e}}_{(i,e)}$ is small-scale fading. we adopt a conservative worst-case model in which an eavesdropper is assumed to experience the same antenna gain as the intended receiver, where $G_{(i,e)} = G_{(i,j)}$. In addition, the legitimate receiver can cancel the coordinate jamming signal through prior randomness, whereas passive Eves cannot. Hence, the jamming term appears only in the wiretap SINR.

The transmission power density of node i is bounded by $P^{\max}_i$ and $P^{\min}_i$, denoted as

$$\rho_i + \sigma_i \le P^{\max}_i, \ \ \rho_i \ge P^{\min}_i \ \forall i \in \mathcal{I}. \tag{3}$$

In the presence of multiple Eves, the secrecy capacity of the link between nodes i and j is defined as

$$C_{(i,j)} = \Big[\log_2\big(1 + \mathrm{SNR}^{\mathsf{s}}_{(i,j)}\big) - \log_2\big(1 + \max_{e \in \mathcal{M}} \mathrm{SNR}^{\mathsf{e}}_{(i,e)}\big)\Big]^+. \tag{4}$$

Then, as introduced in [6], [30]–[32], the SPSC probability between node i and j is defined as $\mathbb{P}_{(i,j)} = \mathbb{P}(C_{(i,j)} > 0)$. For threshold $\tau > 0$, the SPSC constraint can be denoted as

$$\mathbb{P}_{(i,j)} \ge x_{(i,j)} \tau. \tag{5}$$

C. Multi-hop Relay Routing Topology

The routing problem is generally considered as finding a sub-graph of directed graph $\mathcal{G}_{\text{all}} = (\mathcal{N}, \mathcal{E})$ where $\mathcal{N} \subset \mathcal{I} \cup \mathcal{U}$ and $\mathcal{E} \subset \{(i,j) | i, j \in \mathcal{N}\}$. This represents that nodes and users are the graph nodes and the relay hops are the graph edges [20], [21]. The graph follows a spanning tree (ST) topology, which refers to an acyclic graph where all nodes have a single path to the root node [33]. We define $\mathcal{E}_u$ to denote a relay from node 0 to user u, and

$$x_{(i,j)} = \begin{cases} 1, & \text{if } (i,j) \in \mathcal{E} \\ 0, & \text{otherwise} \end{cases}, \ \ x_{(i,j),u} = \begin{cases} 1, & \text{if } (i,j) \in \mathcal{E}_u \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

to indicate whether edge (i,j) belongs to the edge set of graph $\mathcal{G}$ and edge (i,j) belongs to $\mathcal{E}_u$. The root node 0 has no incoming edge, whereas every other node has exactly one:

$$\sum_{i \in \mathcal{N}} x_{(i,0)} = 0, \ \ \sum_{j \in \mathcal{N}} x_{(j,i)} = 1, \forall i \in \mathcal{N} \setminus \{0\}. \tag{7}$$

We introduce a *cut-based* constraint to express that graph $\mathcal{G}$ has a ST topology. Every nonempty subset $S \subseteq \mathcal{N} \setminus \{0\}$ must have at least one selected edge entering from $\mathcal{N} \setminus S$, where

$$\sum_{i \notin S,\, j \in S} x_{(i,j)} \geq 1,\ \forall S \subset \mathcal{N} \setminus \{0\},\ S \neq \emptyset. \tag{8}$$

This constraint prevents disconnected routes by ensuring that every node is reachable from the root through a directed path, thereby enforcing the ST topology.

Relay throughput. The nodes dynamically allocate bandwidth and transmit power on the backhaul link to maximize the network utility. We define $\beta_{(i,j),u}$ to represent the allocated bandwidth of user u at link (i,j); and the spectral efficiency $\gamma_{(i,j)}$ between node i and j as [1]

$$\gamma_{(i,j)} = \log_2\left(1 + \mathrm{SNR}^{\mathsf{s}}_{(i,j)}\right). \tag{9}$$

For user u, let $h_u = \sum_{(i,j)\in\mathcal{E}} x_{(i,j),u}$ denote the number of hops in its route. In the multi-hop routing problem, the end-to-end throughput of user u and the system-wide minimum throughput are determined by the bottleneck link as

$$\min_{u\in\mathcal{U},(i,j)\in\mathcal{E}_u} \frac{\beta_{(i,j),u}\gamma_{(i,j)}}{h_u}. \tag{10}$$

IV. Problem Formulation and Proposed Solution

We aim to max-min throughput across the network, ensuring fair service for users accessing source via multiple hops. Prior literature has extensively explored SPSC probability guarantees in terrestrial multi-hop networks [16], [17], [34], [35]. Similarly, the relay routing that ensures users' QoS while achieving the optimal secure connection probability has also been investigated [36], [37]. The remaining gap is load-coupled multi-user routing under SPSC constraints. In this setting, each route changes the shared-relay loads and routing opportunities of subsequent users.

For brevity of notations, let $\mathbf{B} = \{\beta_{(i,j),u} : i,j \in \mathcal{N}, u \in \mathcal{U}\}$, $\mathbf{P} = \{\rho_i : i \in \mathcal{N}\}$, and $\mathbf{J} = \{\sigma_i : i \in \mathcal{N}\}$. The max-min throughput problem is formulated as

$$\boldsymbol{\mathcal{P}_1}: \max_{\mathcal{G},\mathbf{B},\mathbf{P},\mathbf{J}} \min_{\substack{u\in\mathcal{U},\\(i,j)\in\mathcal{E}_u}} \frac{\beta_{(i,j),u}\gamma_{(i,j)}}{h_u} \tag{$\mathcal{P}_1$a}$$

$$\text{s.t. } \mathbb{P}_{(i,j)} \geq x_{(i,j)}\tau, \tag{$\mathcal{P}_1$b}$$

$$(7),\ (8) \tag{$\mathcal{P}_1$c}$$

$$\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \beta_{(i,j),u} \leq B, \tag{$\mathcal{P}_1$d}$$

$$\rho_i + \sigma_i \leq P_i^{\max}, \tag{$\mathcal{P}_1$e}$$

$$\beta_{(i,j),u} \geq 0, \rho_i \geq P_i^{\min}, \sigma_i \geq 0. \tag{$\mathcal{P}_1$f}$$

Constraint ($\mathcal{P}_1$b) represents the threshold of the secrecy connection probability; ($\mathcal{P}_1$c) is related to the graph topology; ($\mathcal{P}_1$d) is the frequency resource constraint; ($\mathcal{P}_1$e) pertains to the transmission and jamming power constraint; and ($\mathcal{P}_1$f) ensures the positivity of resource variables.

[1] The ergodic spectral efficiency $\mathbb{E}_{|h^{\mathsf{s}}_{(i,j)}|^2}\left[\log_2(1 + \mathrm{SNR}^{\mathsf{s}}_{(i,j)})\right]$ is approximated as $\log_2\left(1 + \rho_i/[n_0 (d^{\mathsf{s}}_{(i,j)})^{\alpha_i}]\right)$. Appendix B validates that this approximation remains accurate over the practical SNR range in this work.

There are two main challenges in Problem $\boldsymbol{\mathcal{P}_1}$. First, the SPSC probability $\mathbb{P}_{(i,j)}$ admits no known closed-form expression. Numerical computation of ($\mathcal{P}_1$b) is computationally impractical, so limiting feasible-set construction under stringent compute budgets. Second, routing and RRM are bidirectionally coupled through ($\mathcal{P}_1$a) and ($\mathcal{P}_1$b). Its joint optimization entails a combinatorial search over routing topologies and their induced RRM [38]. This paper addresses these challenges by deriving a closed-form SPSC approximation and analytically solving the optimal RRM for a given routing, thereby reducing the joint problem to graph optimization.

A. Closed-Form Approximation of the SPSC Probability

For a hop between node i and j, the SPSC probability is defined from (4) as

$$\mathbb{P}_{(i,j)} = \mathbb{P}\left(\log_2\left(\frac{1 + \mathrm{SNR}^{\mathsf{s}}_{(i,j)}}{1 + \max_{e\in\mathcal{M}} \mathrm{SNR}^{\mathsf{e}}_{(i,e)}}\right) > 0\right) \tag{11}$$

$$= \mathbb{P}\left(\mathrm{SNR}^{\mathsf{s}}_{(i,j)} > \max_{e\in\mathcal{M}} \mathrm{SNR}^{\mathsf{e}}_{(i,e)}\right) \tag{12}$$

$$= \mathbb{E}_{|h|^2,\mathcal{M}}\left[\prod_{e\in\mathcal{M}} \mathbb{P}\left(\mathrm{SNR}^{\mathsf{s}}_{(i,j)} > \mathrm{SNR}^{\mathsf{e}}_{(i,e)}\right)\right]. \tag{13}$$

Equation (13) is derived from the property of HPPP and $\mathbb{P}(C > \max\{\gamma_1, ..., \gamma_n\}) = \mathcal{P}(C > \gamma_1, ..., C > \gamma_n)$.

We define $\lambda_i \in \{\lambda_1, \lambda_2, \lambda_3, \lambda_4\}$ as the Eve density of the network layer that node i belongs to. The probability generating functional of HPPP [15] results in $\mathbb{E}_{\mathcal{M}}[\prod_{e\in\mathcal{M}} f(\boldsymbol{p}_e)] = \exp\left(-\lambda_i \int_{\mathbb{R}^2} 1 - f(\boldsymbol{p}_e) d\boldsymbol{p}_e\right)$. Then, $\mathbb{P}_{(i,j)}$ becomes

$$\mathbb{P}_{(i,j)} = \mathbb{E}_{|h|^2}\left[\exp\left[-\lambda_i \int_{\mathbb{R}^2} \mathbb{P}\left(\mathrm{SNR}^{\mathsf{s}}_{(i,j)} < \mathrm{SNR}^{\mathsf{e}}_{(i,e)}\right)\, d\boldsymbol{p}_e\right]\right] \tag{14}$$

$$= \mathbb{E}_{|h|^2}\left[\exp\left[-\lambda_i \int_{\mathbb{R}^2} \mathbb{P}\left(\overbrace{\frac{|h^{\mathsf{s}}_{(i,j)}|^2 (d^{\mathsf{e}}_{(i,e)})^{\alpha_i} n_0}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0 - \sigma_i G_{(i,j)} |h^{\mathsf{s}}_{(i,j)}|^2}}^{\textbf{(a)}} < |h^{\mathsf{e}}_{(i,e)}|^2\right) d\boldsymbol{p}_e\right]\right].$$

From $|h^{\mathsf{e}}_{(i,e)}|^2 \sim \mathsf{Exp}(1)$, which corresponds to unit-mean Rayleigh fading, (14) simplifies to

$$\mathbb{P}_{(i,j)} = \mathbb{E}_{|h|^2}\left[\exp\left[-\lambda_i \underbrace{\int_{\mathbb{R}^2} \exp\left(-\textbf{(a)}\right)\, d\boldsymbol{p}_e}_{\textbf{(b)}}\right]\right]. \tag{15}$$

Direct integral of **(b)** results in

$$\textbf{(b)} = \frac{2\pi}{\alpha_i}\Gamma\left(\frac{2}{\alpha_i}\right)(d^{\mathsf{s}}_{(i,j)})^2 \left(\frac{|h^{\mathsf{s}}_{(i,j)}|^2}{1 - \frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0} |h^{\mathsf{s}}_{(i,j)}|^2}\right)^{-\frac{2}{\alpha_i}}. \tag{16}$$

With $\kappa_i = \lambda_i \frac{2\pi}{\alpha_i}\Gamma(2/\alpha_i)$ and $x = \lambda_i\textbf{(b)}$, integration by parts gives the exact one-dimensional representation

$$\mathbb{P}_{(i,j)}(\sigma_i) = \int_0^\infty \exp\left[-x - \frac{1}{\frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0} + \left(\frac{x}{\kappa_i (d^{\mathsf{s}}_{(i,j)})^2}\right)^{\alpha_i/2}}\right] dx.$$

The factor e^{-x} makes $x = O(1)$ the dominant region of the integral. Thus, the operating condition $[\kappa_i (d^{\mathsf{s}}_{(i,j)})^2]^{-\alpha_i/2} \ll \sigma_i G_{(i,j)}/[(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0]$ requires positive normalized jamming relative to the asymptotically small term. Under this condition, a first-order expansion of the exact integral above gives

$$\mathbb{P}_{(i,j)} = \exp\Big[-\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}{\sigma_i G_{(i,j)}}\Big] \times \Big[1+\Gamma\Big(1+\frac{\alpha_i}{2}\Big)\Big[\kappa_i (d^{\mathsf{s}}_{(i,j)})^2\Big]^{-\frac{\alpha_i}{2}}\Big(\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}{\sigma_i G_{(i,j)}}\Big)^2\Big]. \tag{17}$$

Re-exponentiating the first-order correction yields the closed-form approximation

$$\widehat{\mathbb{P}}_{(i,j)}(\sigma_i) = \exp\Bigg[-\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i}}{\frac{\sigma_i G_{(i,j)}}{n_0} + \Gamma\Big(1+\frac{\alpha_i}{2}\Big)\kappa_i^{-\alpha_i/2}}\Bigg]. \tag{18}$$

A detailed derivation of (18) is provided in Appendix C.

B. Optimal Frequency Resource Allocation

We first look into how the optimal $\mathbf{B}$ is obtained for fixed $\mathcal{G}, \mathbf{P}, \mathbf{J}$. For auxiliary variable η, $\mathcal{P}_1$ can be rewritten as

$$\mathcal{P}_2 : \max_{\mathbf{B},\eta}\ \eta \tag{$\mathcal{P}_2$a}$$

$$\text{s.t. } \beta_{(i,j),u}\gamma_{(i,j)} h_u^{-1} \ge x_{(i,j),u}\eta \tag{$\mathcal{P}_2$b}$$

$$\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \beta_{(i,j),u} \le B. \tag{$\mathcal{P}_2$c}$$

The Lagrangian function of $\mathcal{P}_2$ is

$$\mathcal{L}(\mathbf{B},\eta,\boldsymbol{\lambda},\mu) = \eta + \sum_{\lambda_{(i,j),u}\in\boldsymbol{\lambda}} \lambda_{(i,j),u}\Big(x_{(i,j),u}\eta - \frac{\beta_{(i,j),u}\gamma_{(i,j)}}{h_u}\Big) + \mu\Big(\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\beta_{(i,j),u} - B\Big) \tag{19}$$

for $\boldsymbol{\lambda} = \{\lambda_{(i,j),u} : i,j\in\mathcal{N}, u\in\mathcal{U}\}$. The first-order optimality condition on (19), $\delta\mathcal{L}/\delta\eta = 0,\ \delta\mathcal{L}/\delta\beta_{(i,j),u} = 0$, gives

$$\eta = \Big(\sum \lambda_{(i,j),u} x_{(i,j),u}\Big)^{-\frac{1}{2}},\ \lambda_{(i,j),u} = \mu h_u/\gamma_{(i,j)}. \tag{20}$$

The optimal resource allocation $\beta^*_{(i,j),u}$ occurs when all frequency resources are fully utilized, which corresponds to the equality condition of ($\mathcal{P}_2$c). Otherwise, we have $\mu = \lambda_{(i,j),u} = 0$ and η in (20) is not defined according to complementary slackness of the KKT conditions. Similarly, the equality condition of ($\mathcal{P}_2$b) should be satisfied as well.

This can be expressed mathematically as

$$x_{(i,j),u}\eta - \frac{\beta^*_{(i,j),u}\gamma_{(i,j)}}{h_u} = 0,\ \sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\beta^*_{(i,j),u} = B. \tag{21}$$

Solving the above equations with respect to $\beta^*_{(i,j),u}$ results in

$$\beta^*_{(i,j),u} = B\frac{x_{(i,j),u}h_u}{\gamma_{(i,j)}}\Bigg(\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\frac{x_{(i,j),u}h_u}{\gamma_{(i,j)}}\Bigg)^{-1}. \tag{22}$$

C. Optimal Power Allocation

For fixed $\mathcal{E}$, we can reformulate $\mathcal{P}_1$ by plugging in (22) and introducing an auxiliary variable η as

$$\mathcal{P}_3 : \max_{\mathbf{P},\mathbf{J},\eta}\quad \eta \tag{$\mathcal{P}_3$a}$$

$$\text{s.t. }\ \eta\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\frac{x_{(i,j),u}h_u}{\gamma_{(i,j)}} \le B, \tag{$\mathcal{P}_3$b}$$

$$(\mathcal{P}_1\text{b}),\ (\mathcal{P}_1\text{e}),\ \rho_i \ge P_i^{\min},\ \sigma_i \ge 0. \tag{$\mathcal{P}_3$c}$$

We rigorously prove the following argument by exploring KKT conditions: *The optimal throughput is attained at the maximum transmission power allowed by the system.* We first simplify the constraints and then demonstrate that the maximum achievable transmission power is determined by the minimum jamming power density.

As $d^{\mathsf{s}}_{(i,j)}$ are fixed for given $\mathcal{E}$, the SPSC probability (18) can be viewed as a function of σ_i, denoted as $\mathbb{P}_{(i,j)}(\sigma_i)$. As $\mathbb{P}_{(i,j)}(\sigma)$ is a monotonically increasing function, the constraint ($\mathcal{P}_1$b) can be redefined as

$$\sigma_i \ge \sigma^{\mathsf{req}}_{\tau,(i,j)} = \frac{n_0}{G_{(i,j)}}\Bigg[-\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i}}{\ln\tau} - \Gamma\left(1+\frac{\alpha_i}{2}\right)\kappa_i^{-\alpha_i/2}\Bigg]^+.$$

Thus, σ_i must meet the maximum requirement among node i's outgoing links. Thus, the constraint becomes

$$\sigma_i \ge \tau_i = \max_{j:(i,j)\in\mathcal{E}} x_{(i,j)}\sigma^{\mathsf{req}}_{\tau,(i,j)} \tag{23}$$

Then, Problem $\mathcal{P}_3$ can be simplified as

$$\mathcal{P}_3 : \max_{\mathbf{P},\eta}\quad \eta \tag{$\mathcal{P}_3$a}$$

$$\text{s.t. }\ \eta\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\frac{x_{(i,j),u}h_u}{\gamma_{(i,j)}} \le B, \tag{$\mathcal{P}_3$b}$$

$$\rho_i + \sigma_i \le P_i^{\max},\ \rho_i \ge P_i^{\min},\ \sigma_i \ge \tau_i. \tag{$\mathcal{P}_3$c}$$

With $A_i = \sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\frac{x_{(i,j),u}h_u}{\gamma_{(i,j)}}$, the Lagrangian of $\mathcal{P}_3$ is

$$\mathcal{L}(\mathbf{P},\eta,\boldsymbol{\lambda},\boldsymbol{\mu},\boldsymbol{\nu},\boldsymbol{\xi}) = \eta - \sum_{i\in\mathcal{I}}\lambda_i(B-\eta A_i) - \sum_{i\in\mathcal{I}}\xi_i(\sigma_i-\tau_i) - \sum_{i\in\mathcal{I}}\mu_i(P_i^{\max}-\rho_i-\sigma_i) - \sum_{i\in\mathcal{I}}\nu_i(\rho_i - P_i^{\min}). \tag{24}$$

Then, the derivatives of Lagrangian function are provided as

$$\begin{aligned}&\frac{\partial\mathcal{L}}{\partial\eta} = 1 - \sum_{i\in\mathcal{I}}\lambda_i\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}}\frac{x_{(i,j),u}h_u}{\gamma_{(i,j)}} = 0,\\ &\frac{\partial\mathcal{L}}{\partial\rho_i} = -\lambda_i\eta\frac{\partial A_i}{\partial\rho_i} - \mu_i + \nu_i = 0, \frac{\partial\mathcal{L}}{\partial\sigma_i} = -\mu_i + \xi_i = 0.\end{aligned} \tag{25}$$

Also, the complementary slackness condition indicates $\nu_i = 0$ and $\sigma_i > 0$ since the nodes can transmit signal to other nodes.

Suppose $\sigma_i > \tau_i$ for all $i\in\mathcal{I}$, meaning that we allocate more jamming power than the threshold. Then, we have $\xi_i = \mu_i = 0$ by the complementary slackness condition. Plugging $\mu_i = 0$ into $\frac{\partial\mathcal{L}}{\partial\rho_i}$ yields $\nu_i = -\lambda_i\eta\frac{\partial A_i}{\partial\rho_i} \ge 0$ as $\frac{\partial A_i}{\partial\rho_i} < 0$. If $\nu_i > 0$, then $\rho_i = P_i^{\min}$ by complementary slackness. Then, $\rho_i + \sigma_i < P_i^{\max}$ since $\mu_i = 0$, so decreasing σ_i and increasing ρ_i keeps feasibility, strictly decreases A_i allows a larger η, which is a contradiction. Thus, $\sigma_i = \tau_i$ holds at optimum.

Constraint ($\mathcal{P}_3$c) now can be viewed as $\rho_i \le P_i^{\max} - \tau_i$. If we assume $\rho_i < P_i^{\max} - \tau_i$ for $i\in\mathcal{I}$, then $\mu_i = 0$ to satisfy the complementary slackness condition. Again, we obtain $\lambda_i = 0$ to meet the optimality condition $\frac{\partial\mathcal{L}}{\partial\rho_i}$, which contradicts $\frac{\partial\mathcal{L}}{\partial\eta}$.

Combining the above conditions, the optimal transmission and jamming power, ρ_i^* and σ_i^*, are given as

$$\rho_i^* = P_i^{\max} - \tau_i,\ \sigma_i^* = \tau_i,\ \forall i\in\mathcal{I}, \tag{26}$$

for τ_i in (23). We correspondingly denote the optimal spectral efficiency as $\gamma^*_{(i,j)}$ by putting ρ^*_i into (9).

The following theorem guarantees that the RRM in (22) and (26) is optimal for the given graph $\mathcal{G}$.

Theorem 1 (Optimality of RRM). *Let $\mathcal{G}$ be a fixed feasible network topology. Then, the solution $(\beta^*_{(i,j),u}, \rho^*_i, \sigma^*_i)$ obtained by solving $\mathfrak{P}_1$ under the given $\mathcal{G}$ is globally optimal for the fixed-topology RRM subproblem, provided that $P_i^{\max} - \tau_i \geq P_i^{\min}$ for transmitters with outgoing routed links.*

Proof. The proof is provided in Appendix D. □

D. Load-Aware Dual-Price Routing via Self-Supervised GNN

We propose LOAD to optimize the routing topology under the closed-form optimal RRM. Fig. 2 and Algorithm 1 present the overall flow of LOAD and its routing procedure, respectively. By Theorem 1, the RRM variables are optimally determined for any fixed feasible topology $\mathcal{G}$. After substituting (22) and (26), $\mathfrak{P}_1$ reduces to the routing problem

$$\min_{\mathcal{G}} \max_{i\in\mathcal{I}} L_i(\mathcal{G}) \text{ s.t. } (\mathfrak{P}_1\text{b}), (\mathfrak{P}_1\text{c}), \tag{27}$$

where $L_i(\mathcal{G}) = \frac{1}{B_i}\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \frac{x_{(i,j),u}h_u}{\gamma^*_{(i,j)}}$ is the bandwidth-normalized relay load of transmitter i per unit of common end-to-end throughput. Under the optimal power allocation in (26), achieving a common end-to-end throughput η requires transmitter i to use a fraction $\eta L_i(\mathcal{G})$ of its available bandwidth. Thus, the largest $L_i(\mathcal{G})$ identifies the network bottleneck, and routing seeks a topology that minimizes this maximum load.

To obtain an equivalent formulation without explicit constraints, we characterize the feasible routing domain induced jointly by ($\mathfrak{P}_1$b) and ($\mathfrak{P}_1$c). The SPSC constraint ($\mathfrak{P}_1$b) restricts each outgoing link of node i to

$$d^{\mathsf{s}}_{(i,j)} \leq D^{\mathsf{max}}_{(i,j)}, \ \forall j \in \mathcal{N}, \tag{28}$$

where $D^{\mathsf{max}}_{(i,j)}$ is the farthest allowable distance of link (i,j) under the maximum jamming power density $P_i^{\mathsf{max}} - P_i^{\mathsf{min}}$. Then, routing is restricted to the edge set $\mathcal{E}_\tau$ defined by (28) to meet the SPSC constraint. The ST constraint ($\mathfrak{P}_1$c) then restricts the feasible routing graphs to $\boldsymbol{\mathcal{G}}_\tau$, the set of spanning trees supported on $\mathcal{E}_\tau$. Since every $\mathcal{G} \in \boldsymbol{\mathcal{G}}_\tau$ satisfies both constraints, (27) is equivalently written as

$$\min_{\mathcal{G}\in\boldsymbol{\mathcal{G}}_\tau} \max_{i\in\mathcal{I}} L_i(\mathcal{G}). \tag{29}$$

Dual-price interpretation. For a fixed routing graph $\mathcal{G}$, the minimax load objective $\max_{i\in\mathcal{I}} L_i(\mathcal{G})$ can be reformulated by introducing an auxiliary variable t as

$$\min_t \ t \text{ s.t. } L_i(\mathcal{G}) \leq t, \ \forall i \in \mathcal{I}. \tag{30}$$

Let $\pi_i \geq 0$ be the dual price associated with the load constraint of relay i. The Lagrangian of (30) can be rearranged as

$$\mathcal{L}_{\mathcal{G}}(t, \boldsymbol{\pi}) = \sum_{i\in\mathcal{I}} \pi_i L_i(\mathcal{G}) + \big(1 - \sum_{i\in\mathcal{I}} \pi_i\big)t, \tag{31}$$

We remark that the dual function $\inf_t \mathcal{L}_{\mathcal{G}}(t,\boldsymbol{\pi})$ has a finite value only if $\sum_{i\in\mathcal{I}} \pi_i = 1$. Thus, the dual feasible set is

$$\Delta = \big\{\boldsymbol{\pi} \mid \pi_i \geq 0, \ \sum\nolimits_{i\in\mathcal{I}} \pi_i = 1\big\}. \tag{32}$$

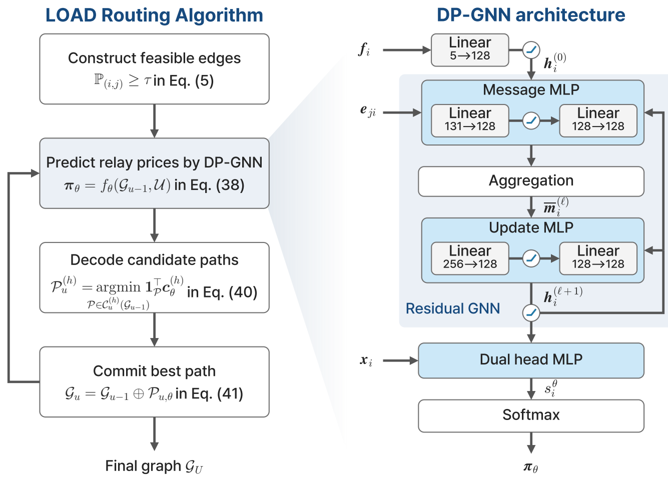


Fig. 2. Flowchart of the LOAD routing algorithm and DP-GNN architecture.

For feasible $\boldsymbol{\pi} \in \Delta$, the lower bound of (31) is $\sum_{i\in\mathcal{I}} \pi_i L_i(\mathcal{G})$. Since problem (30) has a linear objective and constraints, the duality gap is zero, which implies $\max_{i\in\mathcal{I}} L_i(\mathcal{G}) = \max_{\boldsymbol{\pi}\in\Delta} \sum_{i\in\mathcal{I}} \pi_i L_i(\mathcal{G})$. Thus, (29) is equivalent to

$$\min_{\mathcal{G}\in\boldsymbol{\mathcal{G}}_\tau} \max_{\boldsymbol{\pi}\in\Delta} \sum_{i\in\mathcal{I}} \pi_i L_i(\mathcal{G}). \tag{33}$$

Now, we convert (33) into a per-user path selection problem, as directly searching over all feasible graphs remains combinatorial. Let $\mathcal{P}_u$ denote the source-to-user path graph for user $u \in \mathcal{U}$. Then, the routing graph can be expressed as

$$\mathcal{G} = \mathcal{G}_U \text{ for } \mathcal{G}_u = \mathcal{G}_{u-1} \oplus \mathcal{P}_u, \ \forall u \in \mathcal{U} \tag{34}$$

where $\oplus$ denotes graph merging operator. Accordingly, (33) can be represented as the sequential path-selection problem:

$$v(\mathcal{G}_u) = \min_{\{\mathcal{P}_v\}_{u+1}^U} \max_{\boldsymbol{\pi}\in\Delta} \sum_{i\in\mathcal{I}} \pi_i L_i\left(\mathcal{G}_u \oplus \mathcal{P}_{u+1} \oplus \cdots \oplus \mathcal{P}_U\right), \tag{35}$$

$$v(\mathcal{G}_{u-1}) = \min_{\mathcal{P}_u} v\left(\mathcal{G}_{u-1} \oplus \mathcal{P}_u\right) \text{ s.t. } \mathcal{G}_{u-1} \oplus \mathcal{P}_u \in \boldsymbol{\mathcal{G}}_\tau, \tag{36}$$

Finding the exact solution of (36) is computationally intractable since evaluating $v(\mathcal{G}_{u-1} \oplus \mathcal{P}_u)$ requires optimizing over all feasible path combinations of the remaining users $\{u+1, \ldots, U\}$. Therefore, LOAD replaces the exact future-path evaluation with a one-step surrogate derived from the dual representation in (33). Specifically, the cost-to-go of a candidate path is approximated by the dual-weighted load of the post-decision graph:

$$v(\mathcal{G}_{u-1} \oplus \mathcal{P}) \approx \boldsymbol{\pi}_\theta^\top \boldsymbol{L}(\mathcal{G}_{u-1} \oplus \mathcal{P}), \tag{37}$$

where $\boldsymbol{L}(\mathcal{G}_{u-1} \oplus \mathcal{P}) = [L_i(\mathcal{G}_{u-1} \oplus \mathcal{P})]_{i\in\mathcal{I}}$. Here, $\boldsymbol{\pi}_\theta$ is a prospective relay price predicted by the dual-price (DP)-GNN parameterized by θ, replacing the unknown dual price of the completed graph in (33). LOAD predicts this price from the current partial graph as

$$\boldsymbol{\pi}_\theta = f_\theta\left(\mathcal{G}_{u-1}, \mathcal{U}\right), \ \boldsymbol{\pi}_\theta \in \Delta. \tag{38}$$

From the definition of $L_i(\mathcal{G})$, when user u has hop count h, selecting edge (i,j) adds $h/(B_i\gamma^*_{(i,j)})$ to the load of

Algorithm 1 LOAD routing

Require: Nodes $\mathcal{N}$, users $\mathcal{U}$, trained DP-GNN f_θ
Ensure: Routing graph $\mathcal{G}_U$
```
Construct the SPSC-feasible edge set E_τ by (28)
Initialize G_0 ← ({0}, ∅)
for u = 1, ..., U do
    Compute L(G_{u-1}) via (β*_{(i,j),u}, ρ*_i, σ*_i) in (22) and (26)
    Predict the prospective relay price π_θ by DP-GNN
    for each admissible hop count h ∈ H_u do
        Compute c_θ^(h) by (39)
        Decode P_u^(h) by solving (40)
    end for
    Select P_{u,θ} and update G_u by (41)
end for
return G_U
```

transmitter i. Therefore, under the prospective relay price $\boldsymbol{\pi}_\theta$, the weighted load contribution of edge (i,j) is

$$c^{(h)}_{\theta,(i,j)} = \pi_{\theta,i}\frac{h}{B_i\gamma^*_{(i,j)}},\ (i,j)\in\mathcal{E}_\tau. \tag{39}$$

Let $\boldsymbol{c}^{(h)}_\theta = [c^{(h)}_{\theta,(i,j)}]_{(i,j)\in\mathcal{E}_\tau}$ denote the hop-conditioned edge-cost vector. Since the hop count of the selected path is not known until selecting the path, LOAD builds a separate edge-cost vector for each admissible hop count in $\mathcal{H}_u = \{H_{\min}, H_{\min}+1,\ldots\}$, where $H_{\min}$ is the smallest possible hop. For each admissible hop count $h\in\mathcal{H}_u$, we have

$$\mathcal{P}^{(h)}_u = \underset{\mathcal{P}\in\mathcal{C}^{(h)}_u(\mathcal{G}_{u-1})}{\operatorname{argmin}}\ \mathbf{1}^\top_{\mathcal{P}}\boldsymbol{c}^{(h)}_\theta. \tag{40}$$

Here, $\mathcal{C}^{(h)}_u(\mathcal{G}_{u-1})$ denotes the feasible h-hop source-to-user paths whose addition preserves $\boldsymbol{\mathcal{G}}_\tau$, and $\mathbf{1}_{\mathcal{P}}$ is the edge-incidence vector of $\mathcal{P}$ over $\mathcal{E}_\tau$. For each $h\in\mathcal{H}_u$, (40) finds the minimum-cost feasible path with exactly h hops in $\mathcal{O}(h|\mathcal{E}_\tau|)$ time. LOAD then selects the path with the lowest cost across all admissible hop counts, where

$$\mathcal{P}_{u,\theta} = \mathcal{P}^{(h^\star)}_u,\ \mathcal{G}_u = \mathcal{G}_{u-1}\oplus\mathcal{P}_{u,\theta}. \tag{41}$$

DP-GNN architecture. DP-GNN predicts $\boldsymbol{\pi}_\theta$ by considering the current graph $\mathcal{G}_{u-1}$ and remaining users $\{u, u+1,\cdots,U\}$. We formulate the input of DP-GNN as node features $\boldsymbol{x}_i$, edge features $\boldsymbol{e}_{(j,i)}$, and future features $\boldsymbol{f}_i$. Let $\mathcal{N}_\tau(i)$ be the set of neighbor nodes. For $i\in\mathcal{I}$ and $j\in\mathcal{N}_\tau(i)$, the features are

$$\begin{aligned}
\boldsymbol{x}_i &= [i, \mathcal{G}_{u-1}, L_i(\mathcal{G}_{u-1}), B_i, P^{\max}_i, \alpha_i]^\top,\\
\boldsymbol{e}_{(j,i)} &= \Big[\log\big(1+d^{\mathsf{s}}_{(j,i)}\big), 1-\frac{d^{\mathsf{s}}_{(j,i)}}{D^{\mathsf{max}}_{(j,i)}}, \mathbf{1}_{(j,i)\in\mathcal{E}(\mathcal{G}_{u-1})}\Big]^\top,\\
\boldsymbol{f}_i &= [o_i, h_i, \ell_i, d_i, c_i]^\top.
\end{aligned} \tag{42}$$

where o_i, h_i, ℓ_i, d_i, and c_i denote the expected path occupancy, hop mass, load, child-link distance, and serving user count, respectively. These expectations are computed from the minimum-hop and minimum-distance paths obtained using Dijkstra's algorithm to capture the load that potential users $\{u,\ldots,U\}$ may impose on relay i before being connected. The two paths only summarize the demand of future users, while the final routes are determined by the primal decoder.

The feature $\boldsymbol{f}_i$ is first embedded into a 128-dimensional state as $\boldsymbol{h}^{(0)}_i = \mathsf{ReLU}(\boldsymbol{W}_f\boldsymbol{f}_i+\boldsymbol{b}_f)$ and refined by the GNN layers. For $\ell = \{0,\ldots,4\}$, DP-GNN aggregates edge-conditioned neighbor messages and applies a residual update as

$$\begin{aligned}
\bar{\boldsymbol{m}}^{(\ell)}_i &= \frac{1}{|\mathcal{N}_\tau(i)|}\sum_{j\in\mathcal{N}_\tau(i)}\mathsf{MLP}^{(\ell)}_{\mathrm{msg}}\Big(\big[\boldsymbol{h}^{(\ell)}_j \,\|\, \boldsymbol{e}_{(j,i)}\big]\Big),\\
\boldsymbol{h}^{(\ell+1)}_i &= \mathsf{ReLU}\Big(\mathsf{MLP}^{(\ell)}_{\mathrm{upd}}\Big(\big[\boldsymbol{h}^{(\ell)}_i \,\|\, \bar{\boldsymbol{m}}^{(\ell)}_i\big]\Big)+\boldsymbol{h}^{(\ell)}_i\Big).
\end{aligned} \tag{43}$$

For each $i\in\mathcal{I}$, the dual head combines $\boldsymbol{h}^{(5)}_i$ with $\boldsymbol{x}_i$ to produce a logit, which is then normalized across relays by a softmax function, where

$$s^\theta_i = \mathsf{MLP}_{\mathrm{dual}}\Big([\boldsymbol{x}_i \,\|\, \boldsymbol{h}^{(5)}_i]\Big),\ \pi_{\theta,i} = \frac{\exp(s^\theta_i)}{\sum_{k\in\mathcal{I}}\exp(s^\theta_k)}. \tag{44}$$

By construction, the softmax normalization ensures $\pi_{\theta,i}>0$ and $\sum_{i\in\mathcal{I}}\pi_{\theta,i}=1$, such that $\boldsymbol{\pi}_\theta$ automatically satisfies the simplex constraint $\boldsymbol{\pi}_\theta\in\Delta$ in (32).

Self-supervised learning for bottleneck-aligned dual price. The prospective dual price $\boldsymbol{\pi}_\theta$ is not directly available as a training label since the exact price depends on the unknown future routing decisions. However, $\boldsymbol{\pi}_\theta$ affects routing only through the reduced edge costs in (39), which determine the paths in (40). It is therefore sufficient to train the path preference induced by $\boldsymbol{\pi}_\theta$ rather than reproduce the unavailable dual price itself. The closed-form RRM solutions in (22) and (26) provide this supervision by evaluating the bottleneck load of each candidate path without requiring further optimization.

For the current partial graph $\mathcal{G}_{u-1}$, let $\{\mathcal{P}_1,\ldots,\mathcal{P}_K\}\subset\mathcal{C}_u(\mathcal{G}_{u-1})$ denote the candidate paths for user u, and let h_k denote the hop count of $\mathcal{P}_k$. The reduced cost induced by $\boldsymbol{\pi}_\theta$ defines the predicted path preference as

$$p_{\theta,k} = \frac{\exp\big(-\mathbf{1}^\top_{\mathcal{P}_k}\boldsymbol{c}^{(h_k)}_\theta\big)}{\sum_{\ell=1}^K\exp\big(-\mathbf{1}^\top_{\mathcal{P}_\ell}\boldsymbol{c}^{(h_\ell)}_\theta\big)}. \tag{45}$$

The analytical target is constructed from the post-decision bottleneck load as

$$b_k = \|\boldsymbol{L}(\mathcal{G}_{u-1}\oplus\mathcal{P}_k)\|_\infty,\ q_k = \frac{\exp(-b_k)}{\sum_{\ell=1}^K\exp(-b_\ell)}. \tag{46}$$

Thus, $\mathbf{q}$ assigns greater preference to candidate paths that produce smaller bottleneck loads. In implementation, the candidate costs and bottleneck loads are normalized within each candidate set before applying their respective softmax temperatures, as detailed in Appendix E.

DP-GNN is trained to match this analytical path preference and regularize the predicted prices toward a reference distribution via KL divergences, where the training objective is

$$\mathcal{J}(\theta) = D_{\mathrm{KL}}(\mathbf{q}\|\mathbf{p}_\theta)+\lambda D_{\mathrm{KL}}(\boldsymbol{\pi}^{\mathrm{base}}\|\boldsymbol{\pi}_\theta). \tag{47}$$

The first term aligns the path ranking induced by the reduced costs with the analytical bottleneck ranking; and the second term regularizes $\boldsymbol{\pi}_\theta$ toward the reference price distribution $\boldsymbol{\pi}^{\mathrm{base}}$ specified in Appendix E.

V. Numerical Experiments

This section addresses the following research questions:
RQ 1. How accurate is the closed-form approximation of the SPSC probability? ➜ **Section V-B**
RQ 2. How does the max-min throughput in SAGSIN change when security parameters change? ➜ **Section V-D**
RQ 3. How does LOAD perform on real-world position-data-driven testbeds? ➜ **Section V-E**
Following the experimental configuration in Sec. V-A, Secs. V-B-E address RQs 1–3, respectively. All experiments use Python 3.12 on an AMD Ryzen 9 5800X.

A. Experimental Configuration

SAGSIN configurations. Physical layer parameters for SAGSIN are configured as in Table II, referring to [39], [40] and the 3GPP standard [41]. The default Eve density is set as $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (10^{-3}, 2{\cdot}10^{-3}, 3{\cdot}10^{-4}, 10^{-4})$ following the literature [42]. We randomly deploy 150 ground base stations, 150 maritime base stations, 12 HAP stations, and 10 LEO satellites within a geographic area spanning 30° in longitude and 20° in latitude to serve 60 users.

TABLE II
Simulation Parameters

Parameter (Unit)	Ground (G), Maritime (M), HAPs (H)	LEO
Carrier frequency (GHz)	14	20
Total bandwidth (MHz)	250	400
Tx power (dBm)	30	21.5
Tx antenna gain (dBi)	43.2 (G,M,H→LEO) 25 (G,M,H→G,M,H)	38.5
Rx antenna gain (dBi)	39.7 (G,M,H→LEO) 25 (G,M,H→G,M,H)	38.5
Pathloss exponent	2.8 (G), 2.7 (M), 2.6 (H)	2.4

Data-driven testbed setup. We evaluate the framework on position-data-driven testbeds in the Mozambique-Madagascar Channel and Southern North America.[2] Positions of ground, maritime, HAP, and LEO nodes are obtained from OpenCellID, MarineTraffic, Stratocat's Project Loon records [43], and CelesTrak's Starlink ephemerides, respectively. The ground, maritime, and LEO snapshots use 2025-04-22 13:00 UTC, and the HAP snapshots use 2020-09-28 10:00 UTC for Mozambique-Madagascar and 2020-07-29 00:00 UTC for Southern North America.[3]

Baselines. We compare LOAD with the following routing schemes using the RRM formulation in Sec. IV.

- **Exhaustive search:** Serves as the sampling-based reference. It repeatedly constructs candidate routing trees from randomized edge costs, processes users in randomized orders, and reports the best max-min throughput. The number of exhaustive search trials is set to 5,000.
- **Edge-cost MLP/GNN:** Learns positive edge costs from routing-state and link features. The MLP scores edges independently, whereas the GNN first performs message passing over the feasible graph; shortest-path decoding under the learned costs then produces candidates that are committed using the same tree-preserving throughput evaluation.
- **Message passing:** A graph algorithm baseline that samples feasible source-to-user paths and iteratively accepts replacements that improve the analytical throughput. Its detail algorithm and analysis are provided in Appendix G.
- **Hierarchical genetic:** Uses a canonical genetic algorithm with elitism [44] in a hierarchical node-then-edge search [45], [46]. Optional relay nodes are binary-encoded as genes, and each selected-node subgraph is decoded into randomized shortest-path that satisfies ($\mathcal{P}_1$c) and (28). The implementation uses 2,000 generations, population size 50, six mating parents, four kept parents, 5% mutation probability, and early stopping after 100 non-improving generations.
- **Greedy:** Iteratively selects the relay path that maximizes immediate throughput gain of the user.
- **Fixed-metric shortest paths:** Construct the routing tree from a single shortest-path predecessor map under the following fixed edge cost metrics: **Min-distance** uses link distance, **Min-hop** assigns unit cost to every edge, and **Max-spectral-efficiency (Max-SE)** uses link spectral efficiency.

Baseline interpretation. **Greedy** uses LOAD's throughput evaluation but selects each path according to its immediate throughput gain, without accounting for the demand of users that remain unrouted. **Edge-cost MLP/GNN** use parameter counts of the same order as LOAD but omit prospective demand information and the relay-level dual-price structure; the MLP scores edges separately, whereas the GNN conditions its scores on graph context. **Message passing** performs iterative tree-structured load recomputation and multi-user path refinement without a learned predictor; its comparison with LOAD offers insight into the role of learned prediction. Together, these baselines cover complementary routing designs under the same optimal RRM framework. Appendix E details LOAD and the edge-cost MLP/GNN baselines.

B. RQ1: Accuracy of the SPSC Approximation

We validate the closed-form SPSC approximation in two forms directly relevant to the proposed framework. First, we compare the approximated SPSC probability with Monte Carlo estimates over link distance and jamming power. We then examine whether the approximation remains accurate after inversion into the minimum jamming-power requirement used to construct the SPSC-feasible graph. Both experiments use 500,000 Monte Carlo samples.

1) SPSC Accuracy over Distance and Jamming Power: Fig. 3 compares (18) with Monte Carlo evaluations of (5) over jamming powers of $\{-20, -10, 0, 5\}$ dBm. The (λ_i, α_i) pairs are $(10^{-4}, 2.4)$, $(3{\cdot}10^{-4}, 2.6)$, $(2{\cdot}10^{-3}, 2.7)$, and $(10^{-3}, 2.8)$, corresponding to the LEO, HAP, maritime, and ground layers, respectively. We additionally evaluate $(10^{-2}, 3)$ and $(10^{-2}, 4)$ as high-density stress tests.

[2]For Figs. 7-9, the SAGSIN networks are additionally generated over terrain-informed geographic regions at multiple latitudes and longitudes with randomized node positions.

[3]Since no operational datasets for HAP base stations are currently available, the HAP nodes were synthesized with temporal adjustments.

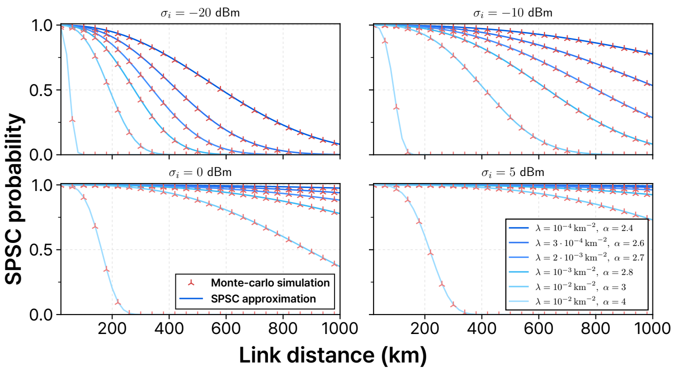


Fig. 3. SPSC probability versus link distance under four jamming-power.

Across all evaluated settings, the maximum absolute difference between the approximation and Monte Carlo estimate is $1.555{\cdot}10^{-3}$. The approximation also preserves the monotonic decrease in SPSC with link distance and the range extension produced by stronger jamming. For example, under $(\lambda_i, \alpha_i) = (10^{-2}, 4)$, the link distances at which the SPSC probability falls to 0.5 are 51, 91, 162, and 217 km for jamming powers of -20, -10, 0, and 5 dBm, respectively. Thus, the approximation reproduces both the probability values and their distance and jamming-power trends, with an absolute error below $1.6{\cdot}10^{-3}$ throughout the tested domain.

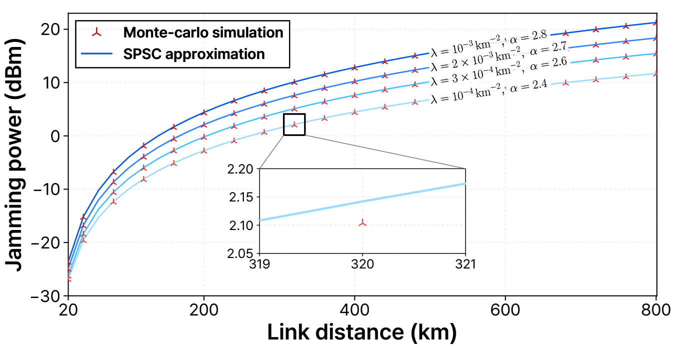


Fig. 4. Minimum jamming power versus link distance for $\tau = 0.999$.

2) Required Jamming-Power Accuracy over Distance: Fig. 4 evaluates the minimum jamming power required to achieve the target SPSC of $\tau = 0.999$ over link distances from 20 to 800 km under the same four nominal SAGSIN layer configurations. The approximated jamming power closely match the Monte Carlo results across all layers and link distances. The maximum absolute difference is 0.052 dB and occurs for the LEO layer at 20 km. At 800 km, the approximation yields required jamming powers of 11.693, 15.458, 18.361, and 21.264 dBm for the LEO, HAP, maritime, and ground layers, respectively, with differences below 0.038 dB from the corresponding Monte Carlo results.

The approximation also reproduces the increase in required jamming power with link distance and the differences among the SAGSIN layers. At every evaluated point, it yields a slightly higher requirement than the Monte Carlo estimate. It therefore does not underestimate the jamming power needed to meet the target SPSC; instead, it includes a small conservative power margin. Consequently, the approximation-based power allocation satisfies the SPSC constraint in a conservative direction, and no additional empirical power correction is required over the tested range.

C. Training Dynamics and Decision Quality

To assess whether training improves routing decisions rather than merely reducing objective values, Fig. 5 reports the learning trajectories of LOAD and the edge-cost MLP/GNN baselines. All three policies reduce their respective losses and stabilize over the 30-epoch schedule. Because LOAD learns dual prices whereas the edge-cost models learn primal edge costs, their losses are not directly comparable; panels (a) and (b) show convergence only within each policy.

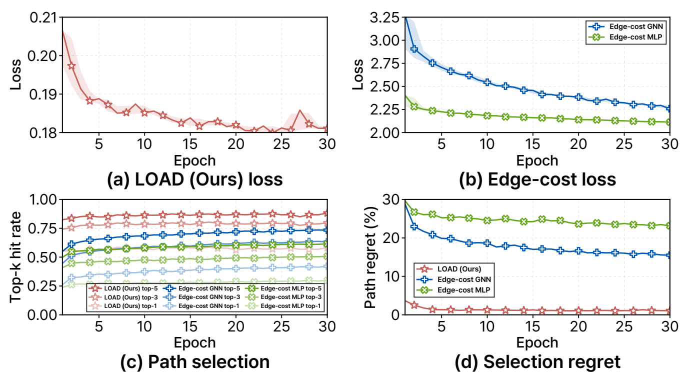


Fig. 5. Training trajectories over 30 epochs for LOAD and edge-cost baselines.

At epoch 30, LOAD attains top-1, top-3, and top-5 hit rates of 58.2%, 79.7%, and 88.2%, respectively, while its hard-selection regret—the committed top-1 path's quality gap from the best candidate in the pool, normalized by the pool's best-to-worst range—is 1.07%. The edge-cost GNN and MLP retain regrets of 15.5% and 23.2%, respectively.

These diagnostics establish the training state of the checkpoints used in the subsequent routing experiments; they are not interpreted as a direct comparison of final routing throughput.

D. RQ2: LOAD Performance under Security Variations

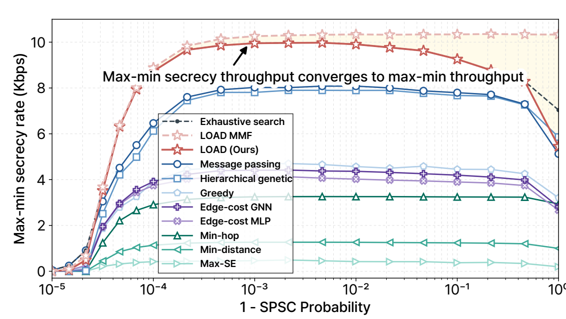


Fig. 6. Max-min throughput and secrecy rate versus SPSC outage probability; the yellow region shows their gap on LOAD-selected routes.

1) Throughput-secrecy gap: Figure 6 compares max-min throughput and secrecy rate on the LOAD-selected routes as the SPSC target varies. This comparison examines whether max-min throughput remains representative of secrecy performance after statistical security has been enforced through

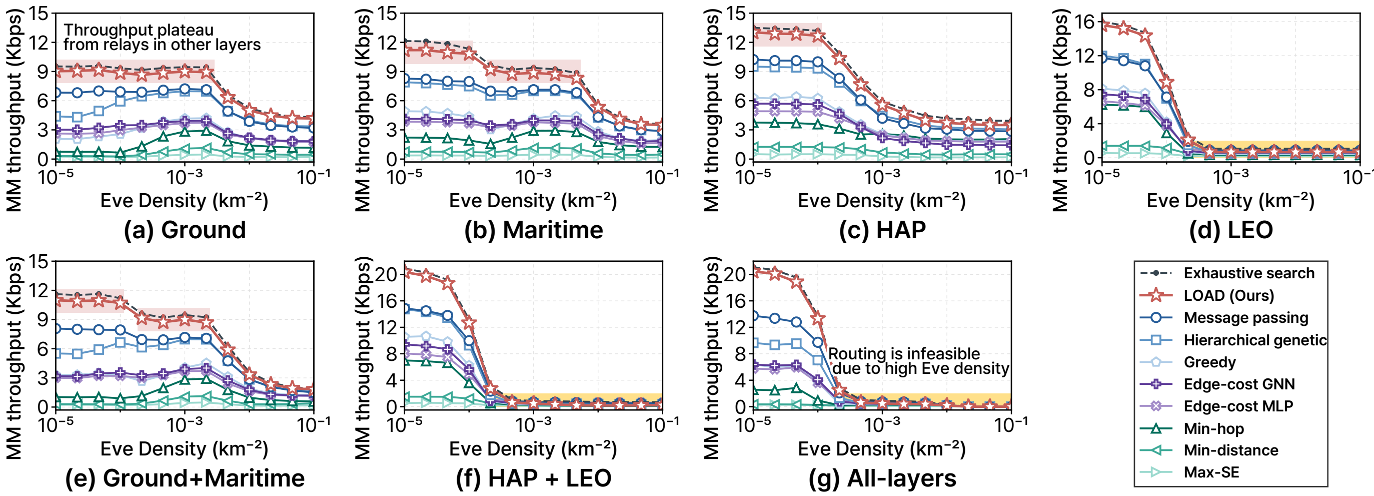
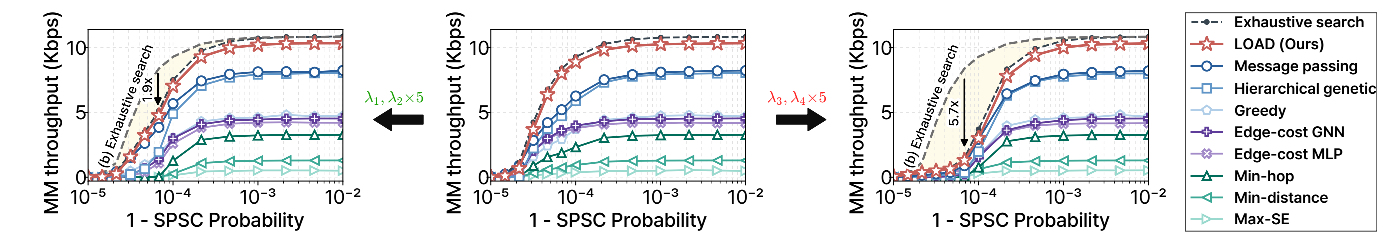


Fig. 7. Max-min throughput versus Eve density for various SAGSIN scenarios. Each graph depicts the max-min throughput while varying the Eve density of the network layer indicated in the caption. The densities of the remaining layers are fixed at their default values.



(a) $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (5{\cdot}10^{-3}, 10^{-3}, 3{\cdot}10^{-4}, 10^{-4})$ **(b)** $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (10^{-3}, 2{\cdot}10^{-3}, 3{\cdot}10^{-4}, 10^{-4})$ **(c)** $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (10^{-3}, 2{\cdot}10^{-3}, 1.5{\cdot}10^{-3}, 5{\cdot}10^{-4})$

Fig. 8. Max-min throughput versus SPSC outage probability. (a), (b), and (c) measure max-min throughput of SAGSINs in the same environmental settings, except for the Eve densities. The gray dashed lines in (a) and (c) represent the numerical results from the **Exhaustive search** reference in (b), illustrating the relative throughput degradation as yellow areas.

the SPSC constraint. Direct max-min secrecy-rate optimization would retain Eve-dependent rate terms throughout routing and RRM, whereas our formulation encodes their effect through SPSC-feasible links and the associated jamming requirements before optimizing throughput. When the SPSC constraint is relaxed, the discrepancy between the two metrics increases because the throughput objective is determined by legitimate-link rates and does not explicitly incorporate the achievable rates at Eves. Meanwhile, a high τ value tightens the feasible-link and jamming requirements, limiting Eve's achievable rate on the selected hops and making secrecy rate approach throughput. The relative gap is only 2.99% at $1-\tau = 10^{-3}$. Since high SPSC targets are the operating regime of interest for mitigating eavesdropping risk [17], this agreement supports max-min throughput as a tractable surrogate for direct max-min secrecy-rate optimization in the high-security regime.

2) Impact of Eve density: Figure 7 shows how the max-min throughput changes when Eve densities in the various SAGSIN layers vary. The SPSC probability threshold τ is fixed at 99.99%. Across all panels, the max-min throughput generally increases as the swept Eve densities decrease. Lower Eve density can increase $D^{\max}_{(i,j)}$ in (28), retain more outgoing edges from the swept layers in $\mathcal{E}_\tau$, and enlarge the relay set over which the bottleneck load in (29) can be reduced.

The curves in Figs. 7a, b, and e plateau at low Eve densities because the feasible edge sets of the ground and maritime layers stop expanding and further reductions provide diminishing RRM gains. Further throughput gains are then constrained by relay loads in the fixed-density layers. Conversely, in Fig. 7g, simultaneous link removal across all layers makes some users unreachable; all 1,000 topology realizations are infeasible at Eve densities of at least $2.15{\cdot}10^{-2}$ km^{-2}. The largest gains in Figs. 7d, 7f, and 7g occur as lower LEO-layer densities restore long-range LEO links, enabling fewer-hop routes with high spectral efficiency and relieving bottleneck relay loads.

3) Impact of the SPSC probability threshold: Figure 8 illustrates how the max-min throughput varies as the SPSC outage probability $(1-\tau)$ increases from 10^{-5} to 10^{-2}. The minimum transmit power P_i for each base station is set to 80% of its total available power. Relative to Fig. 8b, Figs. 8a and 8c increase the Eve densities of ground/maritime and HAP/LEO nodes, respectively, by a factor of five.

As $1-\tau$ approaches zero, the stricter SPSC target reduces the admissible link range in (28), and the minimum throughput approaches zero. All 1,000 LOAD outputs at each plotted threshold connect every user retained in the feasible graph, so the near-zero tail reflects security-induced rate loss rather than a routing failure. Relative to Fig. 8b, increasing the ground/maritime Eve densities in Fig. 8a and the HAP/LEO Eve densities in Fig. 8c reduces throughput. The stronger sensitivity in Fig. 8c is consistent with the role of long-range LoS HAP/LEO links in connecting distant regions with relatively few hops.

At $1-\tau = 10^{-3}$ in Fig. 8b, LOAD maintains the highest mean throughput among the non-exhaustive baselines and outperforms **Message passing** and **Hierarchical genetic**. The

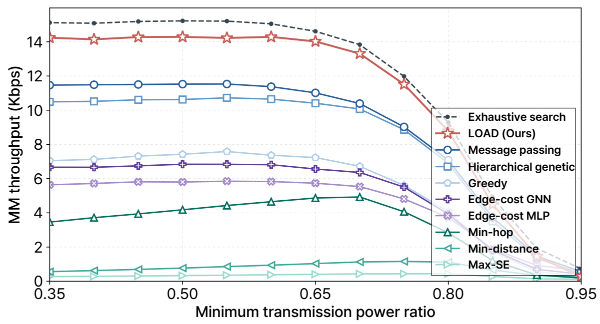


Fig. 9. Max-min throughput versus minimum transmit power ratio. The SPSC probability is set as $\tau = 0.9999$.

separation becomes more evident as a relaxed SPSC constraint restores alternative routes, since throughput then depends on how each scheme distributes future contention across shared relays.

4) Impact of minimum transmit-power constraint: Figure 9 examines how the max-min throughput varies as the minimum transmission power ratio $P_i^{\min}/P_i^{\max}$ changes. For $0.35 \leq P_i^{\min}/P_i^{\max} \leq 0.65$, LOAD remains between 14.0 and $14.3\,$kbps, a variation of 1.9%; the learned and search-based baselines exhibit similar plateaus, whereas the fixed-metric shortest paths vary more substantially. This stable region indicates that the RRM in (26) can satisfy the minimum transmit-power constraint without materially changing the bottleneck allocation.

Beyond this region, a larger mandatory transmit-power allocation leaves less jamming power, reduces the admissible link distance in (28), removes edges from $\mathcal{E}_\tau$, and eventually makes some users unreachable. The fraction with zero throughput for every method rises from 34.95% at $P_i^{\min}/P_i^{\max} = 0.85$ to 64.65% at 0.90 and 80.8% at 0.95. The common failures therefore reflect physical infeasibility under the power split, rather than a routing failure or a computational benefit from reducing the search space.

Cross-setting interpretation. Across the density, SPSC, and power sweeps, the same separation emerges whenever the feasible graph retains several competing routes. LOAD uses prospective relay prices to account for the users that remain unrouted, thereby protecting shared relays before they become throughput bottlenecks. **Message passing** remains the closest comparison because its repeated tree updates can redistribute load after contention appears. **Greedy** commits locally favorable paths without this future-load information, and the edge-cost schemes compress multi-user contention into link scores before shortest-path decoding. The recurring throughput gaps across the three sweeps show that feasible paths alone are insufficient; the routing metric must represent how a current commitment changes the relay capacity available to subsequent users. The inconsistent ordering between **Edge-cost MLP** and **Edge-cost GNN** further shows that graph context alone does not resolve this mismatch when remaining demand and relay-level load are absent from the decoding objective. At high Eve densities or minimum-power ratios, link removal leaves few alternative routes or makes users unreachable, so physical feasibility suppresses these design differences and causes the performance gaps to narrow or occasionally reverse. The repeated ordering links LOAD's throughput gain to anticipatory relay coordination.

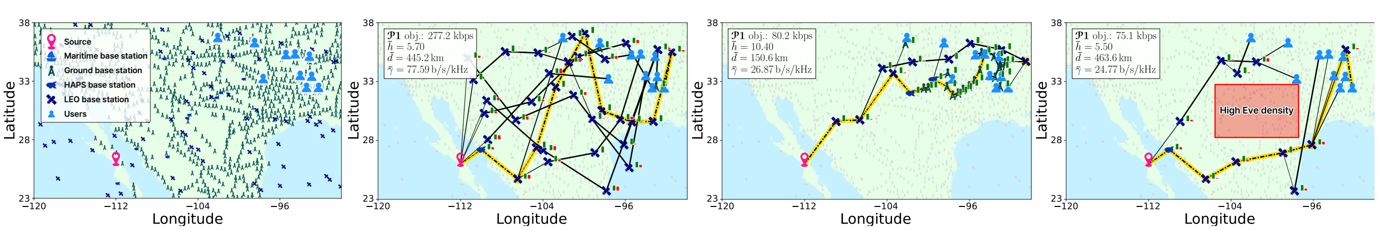
E. RQ3: Routing on Position-Data-Driven Testbeds

To address RQ3, Figs. 10 and 11 examine the routing solutions obtained on the Southern North America and Mozambique–Madagascar testbeds, respectively.

1) Southern North America testbed: Fig. 10 shows how the tested security conditions reshape the feasible graph used by LOAD on the Southern North America testbed. The Eve density and SPSC threshold determine the maximum link distances in (28), while the scenario-specific no-go rules remove additional edges. LOAD then uses the updated topology and remaining-user demand to predict relay prices and decode the feasible paths in Algorithm 1.

Under the moderate condition in Fig. 10b, long-range LEO links remain available. In Fig. 10c, the higher LEO-layer density and the removal of large-swing LEO links restrict LEO connectivity; LOAD then selects routes over the available edges. For the localized condition in Fig. 10d, links intersecting the high-density region are removed during graph construction, so the subsequent decoded routes avoid that region. The panels therefore distinguish security-driven feasibility constraints from relay-price-based route selection.

2) Mozambique-Madagascar Testbed: Fig. 11 compares the secure routing structures on the Mozambique–Madagascar testbed. In Figs. 11b and 11c, LOAD achieves $214.0\,$kbps, only 2.6% below the $219.6\,$kbps achieved by **Exhaustive search**. This small gap is important because max-min throughput is determined by the most heavily loaded relay, not by network-wide averages of hop count, link distance, or spectral efficiency. **Message passing** is the closest scalable baseline, while the locally committed and edge-cost routes leave substantially



(a) Map overview **(b) Moderate Eve density** **(c) High LEO Eve density** **(d) Localized high density**

Fig. 10. Secure routing solutions obtained by LOAD on the Southern North America testbed. In (c), the LEO-layer Eve density is set to $4{\cdot}10^{-4}\,\mathrm{km}^{-2}$ and LEO links spanning more than 4.5° latitude are excluded. In (d), the Eve density in the red region in (a) is $1\,\mathrm{km}^{-2}$ and intersecting links are excluded.

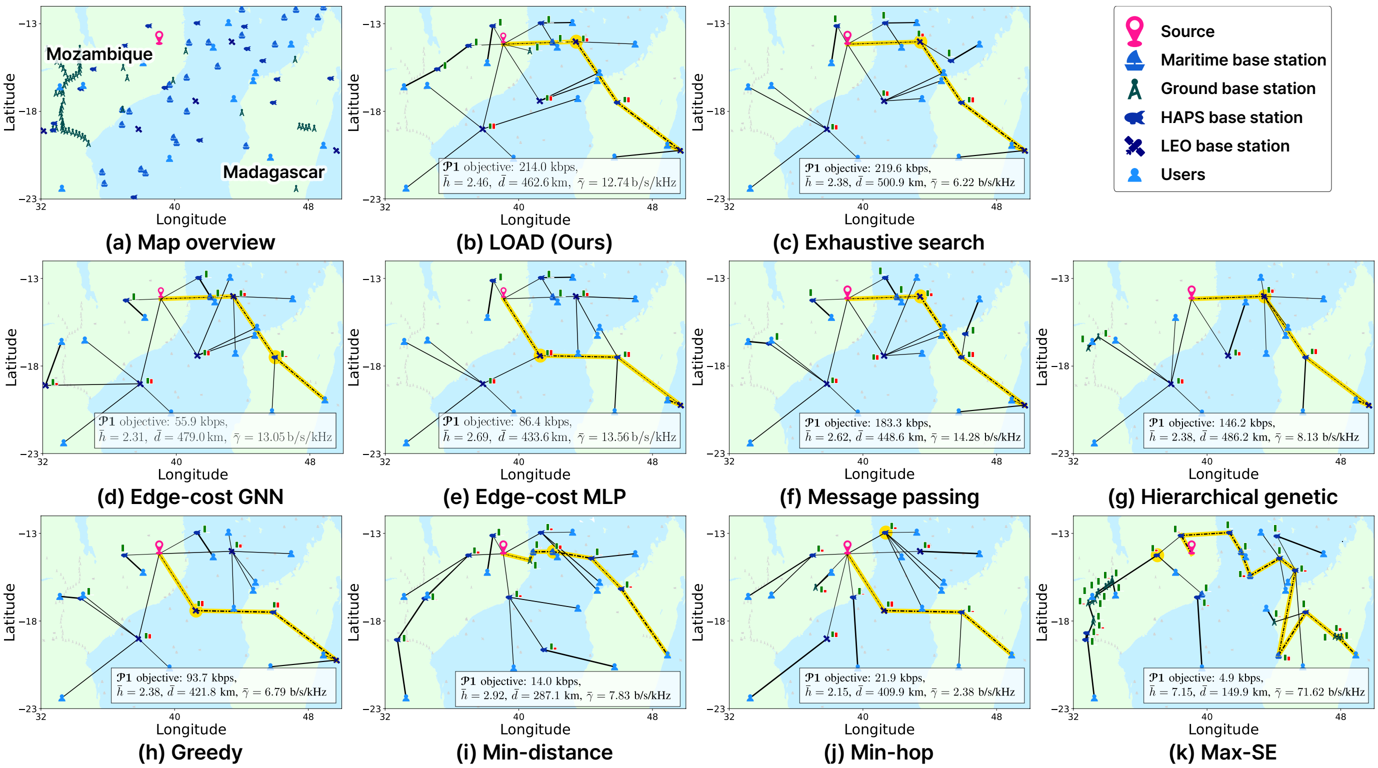


Fig. 11. Secure routing on the Mozambique–Madagascar testbed. The highlighted dashed path and node indicate the longest path and minimum-throughput node, respectively; $\bar{h}$, $\bar{d}$, and $\bar{\gamma}$ denote the average hop count, link distance, and spectral efficiency. The green and red node-adjacent bars show normalized transmission and jamming power, while link width indicates allocated bandwidth.

heavier bottleneck loads. The fixed geography makes the source of this difference explicit: multiple users depend on a small set of long-range relays, so an early path decision changes the capacity available to every downstream choice. Using the future-demand features in (42), LOAD raises the reduced costs in (39) for relays expected to become congested and preserves their capacity for later users. **Message passing** can partly recover this structure through repeated tree updates, but its corrections occur after contention has entered the routing tree. **Greedy** cannot anticipate the effect of its current commitment, and edge-cost decoding does not explicitly propagate future user demand into relay-level congestion. The higher MLP throughput relative to the GNN on this testbed shows that broader graph context is not sufficient unless the learned quantity represents the load-coupled routing objective.

VI. Discussion

Conclusion. This paper developed a cross-layer framework for secure multi-hop relaying in SAGSINs by combining a tractable SPSC approximation, closed-form optimal RRM, and LOAD routing. At $1-\tau = 10^{-3}$, the max-min secrecy rate on LOAD-selected routes was within 2.99% of the corresponding max-min throughput, supporting the throughput objective as a practical surrogate in the evaluated high-security regime. Using prospective relay prices to account for the remaining user demand, LOAD averaged 10.3 kbps across 1,000 topologies at this operating point, compared with 8.1 kbps for **Message passing** and 7.9 kbps for **Hierarchical genetic**. Security sweeps showed how Eve density and the transmit–jamming power split shape feasible links and relay loads. On the position-data-driven testbeds, LOAD reached 214.0 kbps on Mozambique–Madagascar, 2.6% below **Exhaustive search**, and adapted its routes to layer-specific and localized security conditions in Southern North America. Together, these results support combining statistical security modeling, analytical resource allocation, and learned prospective relay prices for secure multi-user routing in SAGSINs.

Future research directions. Prospective prices for shared resources can be combined with model-based incremental costs in other graph-structured network decisions. Potential applications include reconfigurable intelligent surface-assisted connectivity, communication–computation routing in mobile edge computing, and traffic balancing across integrated satellite–terrestrial networks. Future work can address time-varying mobility, traffic, and security through online reconfiguration.

Acknowledgment

We appreciate Prof. Lajos Hanzo for the valuable discussions on dataset construction and for kindly sharing the related research materials on SAGSINs.

References

[1] S. Meng, S. Wu, J. Zhang, J. Cheng, H. Zhou, and Q. Zhang, "Semantics-empowered space-air-ground-sea integrated network: New paradigm, frameworks, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 1, pp. 140–183, 2025.

[2] H. Guo, J. Li, J. Liu, N. Tian, and N. Kato, "A survey on space-air-ground-sea integrated network security in 6G," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 1, pp. 53–87, 2022.

[3] Q. Wang, H. Wang, W. Sun, N. Zhao, H.-N. Dai, and W. Zhang, "Aerial bridge: A secure tunnel against eavesdropping in terrestrial-satellite networks," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 8096–8113, 2023.

[4] Z. Wang, Z. Yin, X. Wang, N. Cheng, Y. Zhang, and T. H. Luan, "Label-free deep learning driven secure access selection in space-air-ground integrated networks," in *IEEE Global Commun. Conf. (GLOBECOM)*, 2023, pp. 958–963.

[5] L. Dong, Z. Han, A. P. Petropulu, and H. V. Poor, "Improving wireless physical layer security via cooperating relays," *IEEE Trans. Signal Processing*, vol. 58, no. 3, pp. 1875–1888, 2010.

[6] F. Jameel, S. Wyne, G. Kaddoum, and T. Q. Duong, "A comprehensive survey on cooperative relaying and jamming strategies for physical layer security," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2734–2771, 2019.

[7] C. Han, L. Bai, T. Bai, and J. Choi, "Joint UAV deployment and power allocation for secure space-air-ground communications," *IEEE Trans. Commun.*, vol. 70, no. 10, pp. 6804–6818, 2022.

[8] Y. Zhang, X. Gao, H. Yuan, K. Yang, J. Kang, P. Wang, and D. Niyato, "Joint UAV trajectory and power allocation with hybrid FSO/RF for secure space–air–ground communications," *IEEE Internet Things J.*, vol. 11, no. 19, pp. 31 407–31 421, 2024.

[9] H. Li, J. Li, M. Liu, and F. Gong, "UAV-assisted secure communication for coordinated satellite-terrestrial networks," *IEEE Commun. Lett.*, vol. 27, no. 7, pp. 1709–1713, 2023.

[10] A. Kakati, G. Li, E. Moustapha Diallo, L. Chiru Kawala, N. Hussain, and A. B. M. Adam, "Toward proactive, secure and efficient space-air-ground communications: Generative AI-based DRL framework," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 1284–1298, 2025.

[11] G. Sun, J. Li, A. Wang, Q. Wu, Z. Sun, and Y. Liu, "Secure and energy-efficient UAV relay communications exploiting collaborative beamforming," *IEEE Trans. Commun.*, vol. 70, no. 8, pp. 5401–5416, 2022.

[12] T. N. Nguyen, T. Van Chien, D.-H. Tran, V.-D. Phan, M. Voznak, S. Chatzinotas, Z. Ding, and H. V. Poor, "Security-reliability tradeoffs for satellite–terrestrial relay networks with a friendly jammer and imperfect CSI," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 7004–7019, 2023.

[13] K. Guo, K. An, B. Zhang, Y. Huang, and D. Guo, "Physical layer security for hybrid satellite terrestrial relay networks with joint relay selection and user scheduling," *IEEE Access*, vol. 6, pp. 55 815–55 827, 2018.

[14] A. S. Abdalla, A. Behfarnia, and V. Marojevic, "UAV trajectory and multi-user beamforming optimization for clustered users against passive eavesdropping attacks with unknown CSI," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 14 426–14 442, 2023.

[15] J. Yao, S. Feng, X. Zhou, and Y. Liu, "Secure routing in multihop wireless ad-hoc networks with decode-and-forward relaying," *IEEE Trans. Commun.*, vol. 64, no. 2, pp. 753–764, 2016.

[16] J. Yao and Y. Liu, "Secrecy rate maximization with outage constraint in multihop relaying networks," *IEEE Commun. Lett.*, vol. 22, no. 2, pp. 304–307, 2018.

[17] J. Yao, X. Zhou, Y. Liu, and S. Feng, "Secure transmission in linear multihop relaying networks," *IEEE Trans. Wireless Commun.*, vol. 17, no. 2, pp. 822–834, 2018.

[18] Y. Xu, J. Liu, Y. Shen, X. Jiang, Y. Ji, and N. Shiratori, "QoS-aware secure routing design for wireless networks with selfish jammers," *IEEE Trans. Wireless Commun.*, vol. 20, no. 8, pp. 4902–4916, 2021.

[19] Y. Xu, J. Liu, Y. Shen, J. Liu, X. Jiang, and T. Taleb, "Incentive jamming-based secure routing in decentralized internet of things," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 3000–3013, 2021.

[20] M. Pagin, T. Zugno, M. Polese, and M. Zorzi, "Resource management for 5G NR integrated access and backhaul: A semi-centralized approach," *IEEE Trans. Wireless Commun.*, vol. 21, no. 2, pp. 753–767, 2022.

[21] H. Yin, S. Roy, and L. Cao, "Routing and resource allocation for IAB multi-hop network in 5G advanced," *IEEE Trans. Commun.*, vol. 70, no. 10, pp. 6704–6717, 2022.

[22] M. H. Eiza and A. Raschellà, "A hybrid SDN-based architecture for secure and QoS aware routing in space-air-ground integrated networks (SAGINs)," in *IEEE Wireless Commun. Networking Conf. (WCNC)*, 2023, pp. 1–6.

[23] H. Luo, Y. Wu, G. Sun, H. Yu, and M. Guizani, "ESCM: An efficient and secure communication mechanism for UAV networks," *IEEE Trans. Netw. Serv. Manag.*, vol. 21, no. 3, pp. 3124–3139, 2024.

[24] M. Sbeiti, N. Goddemeier, D. Behnke, and C. Wietfeld, "PASER: Secure and efficient routing approach for airborne mesh networks," *IEEE Trans. Wireless Commun.*, vol. 15, no. 3, pp. 1950–1964, 2016.

[25] P. Gupta and P. Kumar, "The capacity of wireless networks," *IEEE Trans. Inform. Theory*, vol. 46, no. 2, pp. 388–404, 2000.

[26] S. Goel and R. Negi, "Guaranteeing secrecy using artificial noise," *IEEE Trans. Wireless Commun.*, vol. 7, no. 6, pp. 2180–2189, 2008.

[27] L. Lv, H. Jiang, Z. Ding, L. Yang, and J. Chen, "Secrecy-enhancing design for cooperative downlink and uplink NOMA with an untrusted relay," *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1698–1715, 2020.

[28] B. Chen, R. Li, Q. Ning, K. Lin, C. Han, and V. C. Leung, "Security at physical layer in NOMA relaying networks with cooperative jamming," *IEEE Trans. Veh. Technol.*, vol. 71, no. 4, pp. 3883–3888, 2022.

[29] Y. Mao, H. Zhao, W. Li, and S. Shao, "Joint transceiver optimization for CJ-aided security communication systems with frequency mismatch," *IEEE Wireless Commun. Lett.*, vol. 13, no. 6, pp. 1631–1635, 2024.

[30] L. Kong, S. Vuppala, and G. Kaddoum, "Secrecy analysis of random MIMO wireless networks over α-μ fading channels," *IEEE Trans. Veh. Technol.*, vol. 67, no. 12, pp. 11 654–11 666, 2018.

[31] Y. J. Tolossa, S. Vuppala, G. Kaddoum, and G. Abreu, "On the uplink secrecy capacity analysis in D2D-enabled cellular network," *IEEE Systems J.*, vol. 12, no. 3, pp. 2297–2307, 2018.

[32] N. Bhargav, S. L. Cotton, and D. E. Simmons, "Secrecy capacity analysis over $\kappa - \mu$ fading channels: Theory and applications," *IEEE Trans. Commun.*, vol. 64, no. 7, pp. 3011–3024, 2016.

[33] 3rd Generation Partnership Project (3GPP), "NG-RAN; Architecture description," 3rd Generation Partnership Project (3GPP), Tech. Rep. TS 38.401, March 2023. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.401/

[34] S. Thapar, D. Mishra, and R. Saini, "Novel outage-aware NOMA protocol for secrecy fairness maximization among untrusted users," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13 259–13 272, 2020.

[35] H. Lyu, Y. Kim, and H. J. Yang, "End-to-end secure connection probability in multilayer networks with heterogeneous rician fading," 2026. [Online]. Available: https://arxiv.org/abs/2602.07959

[36] Y. Xu, J. Liu, Y. Shen, X. Jiang, and T. Taleb, "Security/QoS-aware route selection in multi-hop wireless ad hoc networks," in *IEEE Int. Conf. on Commun. (ICC)*, 2016, pp. 1–6.

[37] Y. Xu, J. Liu, O. Takahashi, N. Shiratori, and X. Jiang, "SOQR: Secure optimal QoS routing in wireless ad hoc networks," in *IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2017, pp. 1–6.

[38] H. Lyu, J. Jang, H. Lee, and H. Jong Yang, "Non-iterative optimization of trajectory and radio resource for aerial network," *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 1555–1567, 2025.

[39] D. Liu, J. Zhang, J. Cui, S.-X. Ng, R. G. Maunder, and L. Hanzo, "Deep learning aided routing for space-air-ground integrated networks relying on real satellite, flight, and shipping data," *IEEE Wireless Commun.*, vol. 29, no. 2, pp. 177–184, 2022.

[40] M. Vondra, M. Ozger, D. Schupke, and C. Cavdar, "Integration of satellite and aerial communications for heterogeneous flying vehicles," *IEEE Netw.*, vol. 32, no. 5, pp. 62–69, 2018.

[41] 3rd Generation Partnership Project (3GPP), "Solutions for NR to support Non-Terrestrial Networks (NTN) - Enhancements in Release 18," 3GPP, Technical Report 3GPP TR 38.821, February 2024. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.821/

[42] Y. Zou, J. Zhu, X. Wang, and L. Hanzo, "A survey on wireless security: Technical challenges, recent advances, and future trends," *Proc. IEEE*, vol. 104, no. 9, pp. 1727–1765, 2016.

[43] M. G. Bellemare, S. Candido, P. S. Castro, J. Gong, M. C. Machado, S. Moitra, S. S. Ponda, and Z. Wang, "Autonomous navigation of stratospheric balloons using reinforcement learning," *Nature*, vol. 588, pp. 77–82, 2020.

[44] T. Weise, "Global optimization algorithms-theory and application," *Self-Published Thomas Weise*, vol. 361, 2009.

[45] R. Schaefer and J. Kolodziej, "Genetic search reinforced by the population hierarchy," in *Foundations of Genetic Algorithms 7*. San Francisco, CA, USA: Morgan Kaufmann Publishers, 2003, pp. 383–399.

[46] E. Ciepiela, J. Kocot, L. Siwik, and R. Dreżewski, "Hierarchical approach to evolutionary multi-objective optimization," in *Int. Conf. Comput. Sci. (ICCS)*, M. Bubak, G. D. van Albada, J. Dongarra, and P. M. A. Sloot, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 740–749.